

# NATHAN LEBHERZ

Cell & Data Engineer

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## PROFESSIONAL EXPERIENCE

### Senior Cell Test Engineer

*Tesla, Inc*

Palo Alto, CA

September 2022 - Present

- Lead and first engineer at new lab location overseeing >4000 cell test channels for Li-Ion cell abuse, reliability, formation, characterization and aging test methods. Collaborated across the organization to facilitate construction, equipment bring-up, and operation of new test equipment. Established all cell testing workflows, trained new engineers and technicians, and manage ongoing operations
- Established cell test hardware validation pipeline resulting in 80% time reduction to validate new equipment and development software and firmware releases
- Created shared Github repository of python/SQL scripts to analyze, visualize, and validate cell test data
- Designs and writes cell electrical test schedules including fast charge, formation, electrochemical impedance spectroscopy, AC Ripple, thermal runaway and propagation testing
- Developed strategies for testing electrolyte imbalance, post-formation electrolyte removal, and cell short detection in Lithium-Ion Batteries
- Analyzes battery performance, reliability, runaway risk, and CT scan data for presentations to leadership

### Founder

*Qoindb LLC*

Remote

May 2023 - June 2025

- Solo founder to Qoindb, a custom web app development company designed to serve numismatics
- Engineered a private auction platform with role-based access control (customer/staff/admin), real-time bidding system, and bulk product management generating >\$1.1M in transaction volume
- Developed a comprehensive inventory management system featuring barcode generation/scanning, automated purchase order and invoice creation, and financial analytics dashboards

### Product Engineer

*Clarios Power Solutions, LLC*

Milwaukee, WI

June 2020 – August 2022

- Member of two-year College Graduate Rotational Program providing hands-on lead acid battery experience across Product Testing, Chemical Process Development, Product Design, and Materials
- Executed product deviation testing, risk analysis, and reprocessing strategies to mitigate \$1.4M in on-hold product. Created an automated Python and Power BI dashboard for battery formation analysis to prevent
- Represented the company in collaboration with Argonne National Labs to prototype and test cells for research into cycling characteristics. Formally recognized for timely completion of deliverables
- Assembled and tested 2V battery cells by mixing and hand-pasting electrode recipes
- Deconstructed lead-acid batteries for failure analysis and benchmarking comparisons
- Implemented life-cycling and formation profiles by programming Arbin, MACCOR, and Bitrode cyclers

## EDUCATION

### Bachelor of Science in Chemical Engineering

Minor in Materials Science & Engineering

*University of California, Berkeley*

## TECHNICAL SKILLS

Programming: Python | MATLAB | SQL | Github | JavaScript | HTML | CSS | SolidWorks | COMSOL

Instrumentation & Tools: Arbin | MACCOR | Bitrode | Keyence | SEM/EDS | XRD | Metallography

Core Competencies: Battery Design & Testing (NMC/LFP/PbA), Electrical Protocols (EIS/ACCV/Propagation)

\*References Available\*